SafeHer

Complete Developer Setup Guide

Fresh-machine handover: clone GitHub, configure credentials,
run FastAPI in VS Code, run Android in Android Studio,
and connect the app to the backend.

Repository: https://github.com/Ghige/SAFEHER
Document date: 2026-10-05

NO SECRETS IN THIS DOCUMENT

Source of truth: docs/SAFEHER_DEVELOPER_SETUP.md
Do not use example LAN IPs as permanent values.
Do not commit .env, local.properties, google-services.json, or keystores.

Complete fresh-machine handover for cloning GitHub, configuring credentials, running the FastAPI backend in VS Code, running the Android app in Android Studio, and connecting the two.

- Repository (verified remote): https://github.com/Ghige/SAFEHER.git
- Guide date: 2026-10-05
- Scope: documentation and configuration templates only. This guide does not redesign the app, AI, SOS, auth, APIs, models, or thresholds.
- Companion PDF: docs/SAFEHER_DEVELOPER_SETUP.pdf (regenerate with python scripts/generate_developer_setup_pdf.py)

This Markdown file is the complete source of truth. The PDF is the printable copy of the same guide.

Status labels used in this guide

Table: Label / Meaning

VERIFIED | Confirmed from files in this repository
NOT VERIFIED | Could not be confirmed from the repository; needs a developer/environment check
REQUIRES USER CREDENTIAL | A teammate must provide a private file or secret
REQUIRES EXTERNAL ACCOUNT | Needs Firebase / Google / Meta / Twilio / Postgres access
REQUIRES MANUAL SETUP | Must be done on the machine; Git clone is not enough

1. Project overview

SafeHer is a personal-safety platform:

- Android app (safeher_android/): Kotlin + Jetpack Compose, package com.safeher.app, versionName 2.0.0, versionCode 1 (VERIFIED in app/build.gradle.kts).
- Backend (safeher_backend/): FastAPI app app.main:app, JWT auth, PostgreSQL, optional Redis, Firebase Admin, Meta WhatsApp Cloud API.
- On-device AI: quantized YAMNet TFLite yamnet_distress_quant.tflite bundled in the APK. Live path is confirm-first. AI_AUTO_SOS_ENABLED=false on Android and backend (VERIFIED). Manual SOS is separate.
- Web track page (safeher_web/track/): static emergency-contact map page. Backend can serve it at /track/ when the monorepo layout is intact.
- Training pipeline (safeher_ai/): used to train/export TFLite. Not required to run the app if the production model files are present (they are tracked in Git; VERIFIED via git ls-files).

A developer cloning this repository should not retrain the model just to run the application.

2. Architecture overview (as implemented)

```
Physical phone or emulator
        |
        |  HTTP(S) to FastAPI
        v
safeher_android  ---- Retrofit ---->  safeher_backend (uvicorn :8000)
        |                                   |
        | TFLite in assets                  | PostgreSQL (required)
        | AudioMonitorService (mic FGS)     | Redis (optional in development)
        | LocationForegroundService         | Firebase Admin JSON (FCM)
        | Firebase (google-services.json)   | Meta WhatsApp (optional for local login tests)
        v
Confirm-first AI popup. Auto SOS is OFF.
```

Debug API URL selection (VERIFIED in DebugApiBaseUrl.kt + app/build.gradle.kts):

1. Release builds always use https://api.safeher.app/. DEBUG_API_BASE_URL is compiled as empty. Release forbids cleartext (app/src/main/res/xml/network_security_config.xml).

2. Debug builds with DEBUG_API_BASE_URL set in gitignored local.properties: that LAN URL is used for all debug devices, including emulators.

3. Debug builds with no override:

- Emulator: http://10.0.2.2:8000/ (special alias to the host machine).

- Physical device: Gradle debug BASE_URL also falls back to http://10.0.2.2:8000/ when the override is empty, which does not work on a real phone. For a physical phone you must set DEBUG_API_BASE_URL to your laptop LAN IP.

Debug network security allows cleartext HTTP (app/src/debug/res/xml/network_security_config.xml).

CURRENT MACHINE EXAMPLE ONLY -- DO NOT USE AS A PERMANENT VALUE:

```
http://192.168.1.17:8000/
```

Every developer must look up their own IPv4 on the Wi-Fi that the phone uses. DHCP addresses change.

3. Project structure (actual repository)

```
SAFEHER/
??? README.md                          # Points here and to the PDF
??? PRODUCTION_SETUP.md                # Production DNS / Firebase / WhatsApp checklist
??? TEAM_HANDOVER.pdf                  # Short older onboarding PDF
??? docs/
?   ??? SAFEHER_DEVELOPER_SETUP.md     # This file
?   ??? SAFEHER_DEVELOPER_SETUP.pdf    # Printable copy
??? scripts/
?   ??? generate_developer_setup_pdf.py
?   ??? generate_handover_pdf.py
??? .github/workflows/ci.yml           # Backend pytest on Python 3.11 + Postgres 16
??? safeher_android/                   # Android Studio project root
?   ??? app/                           # Application module
?   ??? gradle/wrapper/                # Gradle 9.3.0 wrapper
?   ??? local.properties.example
?   ??? keystore.properties.example
??? safeher_backend/                   # FastAPI project (open this folder in VS Code)
?   ??? app/                           # Application code (main.py, api, models, services)
?   ??? alembic/versions/              # Migrations 001-008
?   ??? tests/
?   ??? requirements.txt
?   ??? .env.example
?   ??? .env.production.example
?   ??? docker-compose.p0.yml          # Optional Redis + optional Postgres
?   ??? firebase-credentials.json.example
??? safeher_ai/                        # Training / export; models/final holds TFLite
??? safeher_web/                       # Static /track page
??? PLAY_STORE_*.md                    # Store checklists (not required to run locally)
```

Table: Folder / Role

safeher_android/ | Open this directory in Android Studio.
safeher_backend/ | Open this directory in VS Code. Run uvicorn from here.
safeher_ai/ | Train/export YAMNet. Not needed for day-to-day app run.
safeher_web/ | Emergency contact track HTML.
docs/ | This handover.
.github/workflows/ | CI: backend tests with OTP_PROVIDER=mock.

NOT VERIFIED: exact list of every untracked local-only file on a given developer machine.

4. KNOWN SETUP BLOCKERS

Do not treat a Git clone as a one-click install. The following must be completed or acknowledged.

1. REQUIRES USER CREDENTIAL / REQUIRES EXTERNAL ACCOUNT: Android Gradle Google Services plugin is applied (libs.plugins.google.services). A real safeher_android/app/google-services.json is required to sync/build. It is gitignored. Placeholders in google-services.json.example will not talk to Firebase. Obtain the file from a teammate or Firebase Console.

2. REQUIRES USER CREDENTIAL: Backend .env is gitignored. Copy .env.example. PostgreSQL must exist and match DATABASE_URL.

3. REQUIRES MANUAL SETUP: Android local.properties is gitignored. Android Studio usually writes sdk.dir. Physical-device testing also needs DEBUG_API_BASE_URL.

4. REQUIRES EXTERNAL ACCOUNT (optional for login-only local tests): Firebase Admin JSON at FIREBASE_CREDENTIALS_PATH is required for backend FCM. Login/register against Postgres can work without it; push notifications will not. NOT VERIFIED whether every backend boot path fails hard if the JSON is missing.

5. REQUIRES MANUAL SETUP: Physical phone and laptop must share Wi-Fi. Windows Firewall must allow inbound TCP 8000. Uvicorn must bind 0.0.0.0.

6. WhatsApp / Twilio / Maps: Not required to register/login with mock OTP. Required for real SMS OTP, maps tiles, and emergency WhatsApp. REQUIRES EXTERNAL ACCOUNT.

7. GitHub vs this working tree: This documentation describes the workspace as of 2026-10-05, including DEBUG_API_BASE_URL. If a clone of origin/main is missing those files, Gradle/API URL behavior may differ. Check git log / git status after clone. See the final report from the authoring session.

8. Clean-machine install: This guide was not executed on a blank PC as an end-to-end proof. Commands were checked against repository files, not against a fresh OS image.

9. Release signing: keystore.properties and *.jks are gitignored. Debug builds do not need them. Play Store / release APK requires them (REQUIRES USER CREDENTIAL).

10. Training datasets: safeher_ai/data/ is gitignored and not required to run inference.

5. Prerequisites (install first)

REQUIRED

Table: Tool / Version / notes / Status

Git | Any current Git for Windows | VERIFIED needed to clone
Python | 3.11.x (comment in safeher_backend/requirements.txt; CI uses 3.11) | VERIFIED
PostgreSQL | 16 used in CI and docker-compose.p0.yml. Other 14+ versions NOT VERIFIED | VERIFIED for 16 in CI/compose
Android Studio | README previously said Ladybug+. Exact installer version NOT VERIFIED | NOT VERIFIED
JDK | Android compileOptions / kotlinOptions target JVM 17. Android Studio bundled JBR 21 can compile that target. | VERIFIED target 17
Android SDK | Platform 35 (compileSdk = 35, targetSdk = 35), minSdk = 26 | VERIFIED
Gradle | Wrapper 9.3.0 (gradle-wrapper.properties) | VERIFIED
Android Gradle Plugin | 8.7.3 (gradle/libs.versions.toml) | VERIFIED
Kotlin | 2.0.21 | VERIFIED
GitHub access | Clone https://github.com/Ghige/SAFEHER.git (private access may be required) | REQUIRES EXTERNAL ACCOUNT

OPTIONAL

Table: Tool / Why

Redis 7 | Dead-man switch + rate limits. Dev falls back to in-memory if Redis is down (redis_service.py). Production/staging refuse that fallback when REQUIRE_REDIS_IN_PRODUCTION=true.
Docker Desktop | safeher_backend/docker-compose.p0.yml can start Redis and (profile with-db) Postgres.
Meta WhatsApp Cloud API | Emergency contact messages.
Twilio Verify | Real SMS OTP (OTP_PROVIDER=twilio).

Google Maps API key | Map tiles. Empty MAPS_API_KEY still builds.
VS Code Python extension | Convenience only.

DEVELOPMENT ONLY

- OTP_PROVIDER=mock and OTP_ALLOW_DEV_SHORTCUT=true (defaults in .env.example).
- Debug cleartext HTTP to LAN / 10.0.2.2.
- ENVIRONMENT=development, DEBUG=true.

PRODUCTION ONLY (do not use for this local guide)

- ENVIRONMENT=production, OTP_PROVIDER=twilio, OTP_ALLOW_DEV_SHORTCUT=false.
- Release base URL https://api.safeher.app/.
- Redis required. See PRODUCTION_SETUP.md and .env.production.example.

SDK components (VERIFIED as required by Gradle; exact SDK Manager checkbox names NOT VERIFIED):

- Android SDK Platform 35
- Android SDK Build-Tools (version not pinned in Gradle; Android Studio installs a default)
- Android SDK Platform-Tools (adb)
- Google Play services / Google APIs as pulled by Gradle dependencies
- Emulator image: x86_64 or ARM system image with Google APIs, API 26+ (API 35 recommended). Exact AVD name NOT VERIFIED.

6. Developer credential checklist

REQUIRED before a useful local run

- GitHub access to this repository
- Python 3.11
- PostgreSQL running with a database matching DATABASE_URL
- Backend .env copied from .env.example (placeholders replaced)
- SECRET_KEY changed from the example string for anything beyond throwaway local use
- Android Studio + JDK 17-capable toolchain
- Android SDK Platform 35
- safeher_android/local.properties with valid sdk.dir (Studio usually creates this)
- Real safeher_android/app/google-services.json (not the .example file)
- Emulator or physical Android 8.0+ (API 26+) device
- Runtime grants: microphone, notifications, location (when using SOS/maps/AI)

REQUIRED for physical-phone + backend

- Laptop and phone on the same Wi-Fi
- Laptop LAN IPv4 (not 10.0.2.2, not 127.0.0.1)
- DEBUG_API_BASE_URL=http://<YOUR_LAN_IP>:8000/ in local.properties
- Uvicorn --host 0.0.0.0 --port 8000
- Windows Firewall inbound TCP 8000 allowed

REQUIRED for FCM / backend push

- Firebase project access
- Backend firebase-credentials.json (Admin SDK) and FIREBASE_CREDENTIALS_PATH

OPTIONAL

- Redis
- MAPS_API_KEY

- Meta WhatsApp tokens
- Twilio Verify credentials
- Release keystore (keystore.properties + JKS)

PRODUCTION ONLY (not this guide)

- Production .env from a secret store
- Approved WhatsApp templates
- DNS for api.safeher.app / safeher.app

7. SAFE SECURITY RULES

NEVER COMMIT

- Passwords and database URLs with real passwords
- API keys (OpenAI, Google, Maps, Firebase client keys, Meta, Twilio)
- Private keys and PEM files
- Firebase Admin / service-account JSON (firebase-credentials.json, *adminsdk*.json, *service-account*.json)
- Production database credentials
- JWT SECRET_KEY values
- Filled .env files
- local.properties (machine SDK path + optional LAN URL + Maps key)
- google-services.json (contains a Google API key; gitignored)
- Signing keystores (*.jks, *.keystore) and keystore.properties
- User recordings / safeher_ai/data/ datasets

Root .gitignore already ignores .env, local.properties, google-services.json, firebase-credentials.json, venvs, build/, and safeher_ai/data/.

DO COMMIT (templates only)

- safeher_backend/.env.example
- safeher_backend/.env.production.example
- safeher_backend/firebase-credentials.json.example
- safeher_android/local.properties.example
- safeher_android/keystore.properties.example
- safeher_android/app/google-services.json.example

Never paste real secret values into this Markdown, the PDF, README, or chat logs.

8. Configuration files table

Table: File / Required? / Used by / Where to put it / How to obtain it / Commit to Git?

File: safeher_backend/.env -- Required?: Yes (runtime) -- Used by: Backend -- Where to put it: safeher_backend/.env -- How to obtain it: Copy .env.example; fill locally -- Commit to Git?: NO

File: safeher_backend/.env.example -- Required?: Template -- Used by: Humans / this guide -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES

File: safeher_backend/.env.production.example -- Required?: Template -- Used by: Production ops -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES

File: safeher_backend/firebase-credentials.json -- Required?: Required for FCM; optional for login-only -- Used by: Firebase Admin -- Where to put it: Path in FIREBASE_CREDENTIALS_PATH (default ./firebase-credentials.json) -- How to obtain it: Firebase Console: Project settings > Service accounts > Generate new private key -- Commit to Git?: NO

File: safeher_backend/firebase-credentials.json.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES

File: safeher_android/local.properties -- Required?: Yes -- Used by: Gradle -- Where to put it: safeher_android/local.properties --

How to obtain it: Copy example; Studio writes sdk.dir -- Commit to Git?: NO
File: safeher_android/local.properties.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_android/app/google-services.json -- Required?: Yes (Gradle Google Services + FCM) -- Used by: Android -- Where to put it: safeher_android/app/google-services.json -- How to obtain it: Firebase Console download, or teammate via a private channel -- Commit to Git?: NO
File: safeher_android/app/google-services.json.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_android/keystore.properties -- Required?: Release only -- Used by: Gradle signing -- Where to put it: safeher_android/keystore.properties -- How to obtain it: Copy example; private keystore -- Commit to Git?: NO
File: yamnet_distress_quant.tflite -- Required?: Yes (on-device AI) -- Used by: Android assets / Gradle syncTfliteAsset -- Where to put it: Tracked copies under assets, safeher_ai/models/final/, safeher_backend/app/ai/audio_model/ -- How to obtain it: Already in Git -- Commit to Git?: YES (model binary, not a secret)
File: yamnet_distress.tflite -- Required?: Optional (non-quant) -- Used by: Training artifacts -- Where to put it: safeher_ai/models/final/ -- How to obtain it: In Git -- Commit to Git?: YES
File: Alembic versions 001-008 -- Required?: Yes -- Used by: DB schema -- Where to put it: safeher_backend/alembic/versions/ -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_backend/docker-compose.p0.yml -- Required?: Optional -- Used by: Redis / Postgres -- Where to put it: safeher_backend/ -- How to obtain it: In Git -- Commit to Git?: YES
File: CI secrets -- Required?: CI only -- Used by: GitHub Actions -- Where to put it: Workflow env in .github/workflows/ci.yml uses dummy SECRET_KEY and test DB -- How to obtain it: In Git (dummy values) -- Commit to Git?: YES (dummy only)

google-services.json is not safe to publish on GitHub in this project: it is gitignored and contains a restricted Google API key. Share it privately. The .example file is placeholders only.

Observed local Firebase project id (from a private local file, not committed): safeher-4521a. Teammates need Console access to that project or their own Firebase project with matching Android application ids. REQUIRES EXTERNAL ACCOUNT.

oauth_client arrays in the observed local google-services.json were empty. SHA-1 / SHA-256 for Google Sign-In are NOT VERIFIED as required for the current phone+password/OTP login path.

9. Backend setup (VS Code)

9.1 Open the project

1. Install VS Code.
2. File > Open Folder > select safeher_backend (not the monorepo root, unless you prefer a multi-root workspace).
3. Select Python 3.11 as the interpreter after the venv exists.

9.2 Virtual environment (Windows PowerShell)

Commands below match README / CI style. README uses .venv inside safeher_backend. Either .venv or venv311 is fine; this guide uses .venv as the documented default.

```
cd safeher_backend
py -3.11 -m venv .venv
.\.venv\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt
```

If py -3.11 is missing, install Python 3.11 from python.org and retry. Wrong major versions are a common ModuleNotFoundError / wheel source.

requirements.txt pins FastAPI 0.115.6, uvicorn 0.34.0, SQLAlchemy 2.0.36, Alembic 1.14.0, pytest 8.3.4, firebase-admin, redis, psycopg2-binary, numpy 1.26.4.

9.3 Configure .env

```
copy .env.example .env
```

Then edit .env. Do not invent production passwords. Placeholders from .env.example:

```
DATABASE_URL=postgresql://postgres:password@localhost:5432/safeher_db
SECRET_KEY=change-this-to-a-random-secret-key
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=1440
REDIS_URL=redis://localhost:6379
ENVIRONMENT=development
DEBUG=true
OTP_PROVIDER=mock
OTP_ALLOW_DEV_SHORTCUT=true
AI_AUTO_SOS_ENABLED=false
FIREBASE_CREDENTIALS_PATH=./firebase-credentials.json
CORS_ALLOWED_ORIGINS=
```

- DATABASE_URL user/password/db must match the Postgres role you actually created. The example uses user postgres, password password, db safeher_db, host localhost, port 5432. Those are template values, not a guarantee your machine uses them.
- Optional Docker compose Postgres uses user safeher, password safeher_dev_only_change_me, db safeher_db (see section 10).
- Development CORS is * when ENVIRONMENT=development (VERIFIED in app/main.py).
- TrustedHostMiddleware is off unless production-like (_PROD).

9.4 Firebase Admin (backend)

1. Firebase Console > Project settings > Service accounts > Generate new private key.
2. Save as safeher_backend/firebase-credentials.json (gitignored).
3. Leave FIREBASE_CREDENTIALS_PATH=./firebase-credentials.json or set an absolute path.
4. Do not commit this file. The .example file is a shape-only placeholder.

9.5 Start the API

Local-only testing (browser/Swagger on the same PC; Android emulator can still use 10.0.2.2 which maps to the host). Default uvicorn bind without --host is 127.0.0.1:

```
python -m uvicorn app.main:app --reload --port 8000
```

Physical Android device testing (phone must reach the laptop). Bind all interfaces:

```
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

README also documents:

```
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

(equivalent if the venv Scripts directory is on PATH).

Verify Swagger

- Open http://127.0.0.1:8000/docs
- Liveness: GET http://127.0.0.1:8000/api/v1/ping/ should return "status": "ok" and "ai_auto_sos_enabled": false
- Readiness: GET http://127.0.0.1:8000/api/v1/ping/ready requires PostgreSQL (database=ok or HTTP 503)

Do not send a real emergency alert while verifying setup.

10. Database setup (PostgreSQL)

Alembic reads DATABASE_URL from settings (alembic/env.py). alembic.ini has an empty sqlalchemy.url; do not

put passwords there.

There is no automatic user seed. Register a user through the app or POST /api/v1/auth/register. Model-version seed exists only for AI dataset admin endpoints (ensure_seed_model_versions) and is not required for login.

Option A -- local PostgreSQL (manual)

Install PostgreSQL 16 (version 16 VERIFIED in CI). In psql or pgAdmin, create a database. Example using the template credentials from .env.example (replace if your install uses a different postgres password):

```
CREATE DATABASE safeher_db;
```

If you create a dedicated user (optional; do not invent production passwords):

```
CREATE USER safeher WITH PASSWORD '<YOUR_PASSWORD>';
GRANT ALL PRIVILEGES ON DATABASE safeher_db TO safeher;
```

Then set:

```
DATABASE_URL=postgresql://<YOUR_USER>:<YOUR_PASSWORD>@localhost:5432/safeher_db
```

Windows default superuser is often postgres. The password is whatever you chose at install time (REQUIRES MANUAL SETUP).

Option B -- Docker (optional)

From safeher_backend:

```
docker compose -f docker-compose.p0.yml --profile with-db up -d
```

Compose values (VERIFIED in docker-compose.p0.yml):

- user: safeher
- password: safeher_dev_only_change_me (dev-only sample in compose; change if you expose the port)
- database: safeher_db
- port: 5432

Matching URL:

```
DATABASE_URL=postgresql://safeher:safeher_dev_only_change_me@localhost:5432/safeher_db
```

Redis-only:

```
docker compose -f docker-compose.p0.yml up -d
```

Migrations (required)

From safeher_backend with venv active and .env loaded:

```
alembic upgrade head
```

Tracked revisions (VERIFIED filenames):

Table: Revision file / Purpose

001_create_users.py | users
002_create_alerts.py | alerts / contacts related schema
003_create_helpers.py | helpers
004_add_user_is_verified.py | is_verified
005_add_user_fcm_token.py | FCM token column
006_ai_dataset_and_model_versions.py | AI dataset tables
007_uniqueness_alerts_contacts.py | uniqueness constraints
008_ai_feedback_events.py | AI feedback events

Exact live column lists: NOT VERIFIED in this guide beyond migration filenames; inspect the migration files if you need a schema dump.

CI uses database safeher_test with user/password safeher -- that is for GitHub Actions only.

11. Firebase / Google configuration

Android application ids (VERIFIED)

Table: Variant / Application id
Release | com.safeher.app
Debug | com.safeher.app.debug (applicationIdSuffix = ".debug")

Register both Android apps in Firebase if you run debug installs.

google-services.json

- Required at safeher_android/app/google-services.json
- Used by the Google Services Gradle plugin and Firebase (FCM, Crashlytics plugin is applied)
- Do not commit
- Download from Firebase Console for this project, or receive it through a private team channel
- Copying google-services.json.example to google-services.json is not sufficient for a working Firebase connection

Backend Admin SDK

- File: firebase-credentials.json (private key -- never commit)
- Env: FIREBASE_CREDENTIALS_PATH
- Needed for server-side FCM

Auth in this codebase

App login is phone + password via FastAPI (/api/v1/auth/register, /login, /send-otp, /verify-otp). It is not Firebase Auth email/password. Firebase on Android is used for messaging (and Crashlytics), not as the primary user table.

SHA-1 / SHA-256

NOT VERIFIED as required for current OTP/password login. Add debug/release SHA fingerprints in Firebase if you later enable Google Sign-In or restricted API keys that demand them.

Google Maps

MAPS_API_KEY in local.properties is injected as BuildConfig.MAPS_API_KEY and a manifest placeholder. Empty key: app still builds; maps tiles fail.

12. Android setup (Android Studio)

12.1 Open the project

Open the folder safeher_android, not the monorepo root (unless you add it as a module later). Wait for Gradle sync.

12.2 local.properties

```
cd safeher_android
copy local.properties.example local.properties
```

Set:

```
sdk.dir=C:\\Users\\<YourUser>\\AppData\\Local\\Android\\Sdk
MAPS_API_KEY=<YOUR_KEY_OR_LEAVE_EMPTY>
```

Android Studio usually overwrites/creates sdk.dir automatically.

Physical device extra line:

```
DEBUG_API_BASE_URL=http://<YOUR_LAPTOP_LAN_IPV4>:8000/
```

CURRENT MACHINE EXAMPLE ONLY -- DO NOT USE AS A PERMANENT VALUE: http://192.168.1.17:8000/

Find your IPv4 (Windows PowerShell):

```
Get-NetIPAddress -AddressFamily IPv4 | Where-Object { $_.PrefixOrigin -ne 'WellKnown' } | Fo
rmat-Table IPAddress, InterfaceAlias
```

Use the Wi-Fi adapter address (often 192.168.x.x or 10.x.x.x). Do not use 127.0.0.1 (that is the phone itself). Do not use 10.0.2.2 on a physical phone (that alias exists only inside the Android emulator).

After changing DEBUG_API_BASE_URL, Rebuild the app so BuildConfig regenerates.

12.3 google-services.json

Place the real file at safeher_android/app/google-services.json. Gradle sync fails or Firebase init fails without it.

12.4 Sync / build

In Android Studio: File > Sync Project with Gradle Files, then Run.

CLI (from safeher_android, JDK on PATH):

```
.\gradlew.bat assembleDebug
```

Unit tests:

```
.\gradlew.bat testDebugUnitTest
```

JAVA_HOME should point at JDK 17+ (Android Studio JBR is acceptable). NOT VERIFIED that every machine already has JAVA_HOME set.

12.5 Permissions (VERIFIED in AndroidManifest.xml)

Install-time / declared:

- INTERNET, ACCESS_NETWORK_STATE
- ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, ACCESS_BACKGROUND_LOCATION
- RECORD_AUDIO
- POST_NOTIFICATIONS
- FOREGROUND_SERVICE, FOREGROUND_SERVICE_LOCATION, FOREGROUND_SERVICE_MICROPHONE
- REQUEST_IGNORE_BATTERY_OPTIMIZATIONS

Runtime (grant on device): microphone, notifications (API 33+), location (and background location if the OS prompts for SOS tracking).

Foreground services:

- LocationForegroundService -- foregroundServiceType=location
- AudioMonitorService -- foregroundServiceType=microphone

12.6 Emulator vs physical device

See sections 16--18 of the run procedure.

USB debugging on a physical phone: Settings > Developer options > USB debugging. Authorize the PC RSA prompt. adb devices should show device, not unauthorized or offline.

13. AI / SafeHer model setup (as implemented)

Table: Item / Value / Status

Production on-device model | yamnet_distress_quant.tflite | VERIFIED
Android asset name | AiModelAssets.AUDIO_MODEL | VERIFIED
Format | TensorFlow Lite (quantized) | VERIFIED
Bundled in APK? | Yes. Gradle copies into app/src/main/assets/ if missing (syncTfliteAsset) | VERIFIED
Downloaded at runtime? | No | VERIFIED

Retrain required to run app? | No | VERIFIED
Training dataset required for inference? | No (safeher_ai/data/ gitignored) | VERIFIED
Sample rate / window | 16 kHz mono, 1 second / 16000 samples (implementation in audio engine) | VERIFIED in Stage 3 audio pipeline comments/code
Movement TFLite | Not on-device (movement_model.tflite not a required RC1 asset) | VERIFIED
Speech sklearn | Not on-device; phrase catalog + optional Android SpeechRecognizer | VERIFIED
Auto SOS | AI_AUTO_SOS_ENABLED=false (Android BuildConfig + backend settings) | VERIFIED
Confirm-first | Backend CONFIRMATION_WINDOW_SECONDS=30; Android confirm-first popup | VERIFIED window setting; UI path exists in app services
Backend copy of model | safeher_backend/app/ai/audio_model/yamnet_distress_quant.tflite | VERIFIED tracked
Non-quant sibling | safeher_ai/models/final/yamnet_distress.tflite | VERIFIED tracked

Python training extras (safeher_ai/requirements_training.txt) are optional. Backend numpy is pinned for server-side helpers; on-device inference uses Android TensorFlow Lite 2.16.1 (libs.versions.toml).

Do not replace the production TFLite as part of setup.

Logcat filter for live AI (if Stage 3A is present in the checked-out code): tag SAFEHER_AI. If that tag is absent on an older commit, use AudioMonitorService / TfliteAudioInferenceEngine.

14. Complete run procedure (fresh machine)

STEP 1 -- Install prerequisites

Install Git, Python 3.11, PostgreSQL 16, Android Studio (SDK 35). Optional: Redis, Docker Desktop.

STEP 2 -- Clone repository

```
git clone https://github.com/Ghige/SAFEHER.git
cd SAFEHER
```

If the repository is private, authenticate with GitHub (browser, PAT, or SSH). NOT VERIFIED which auth method your org requires.

STEP 3 -- Open backend in VS Code

Open folder SAFEHER\safeher_backend.

STEP 4 -- Create Python virtual environment

```
py -3.11 -m venv .venv
.\.venv\Scripts\activate
```

STEP 5 -- Install dependencies

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

STEP 6 -- Configure .env

```
copy .env.example .env
```

Set DATABASE_URL and SECRET_KEY. Keep AI_AUTO_SOS_ENABLED=false, OTP_PROVIDER=mock, OTP_ALLOW_DEV_SHORTCUT=true for local mock OTP.

STEP 7 -- Configure PostgreSQL

Create database safeher_db (or match your URL). Then:

```
alembic upgrade head
```

STEP 8 -- Configure Firebase / backend credentials

Place Admin SDK JSON if you need FCM. Skip only if you accept no push from backend.

STEP 9 -- Start backend

Emulator / same-PC:

```
python -m uvicorn app.main:app --reload --port 8000
```

Physical phone:

```
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

STEP 10 -- Verify Swagger

Browse http://127.0.0.1:8000/docs and http://127.0.0.1:8000/api/v1/ping/.

STEP 11 -- Open Android project in Android Studio

Open SAFEHER\safeher_android.

STEP 12 -- Configure local.properties

Copy the example; confirm sdk.dir. For a physical phone, set DEBUG_API_BASE_URL to your LAN URL.

STEP 13 -- Configure google-services.json

Place the real file in safeher_android/app/.

STEP 14 -- Sync Gradle

Android Studio: Sync Project with Gradle Files.

STEP 15 -- Build Android project

Run app (debug) or .\gradlew.bat assembleDebug.

STEP 16 -- Run emulator

Create an AVD (API 26+, Google APIs). Cold boot. Run the app configuration. Default debug URL http://10.0.2.2:8000/ reaches uvicorn on the PC if DEBUG_API_BASE_URL is unset. If you set a LAN override, the emulator will also use that LAN IP; the PC must listen on 0.0.0.0.

STEP 17 -- Run physical Android device

Enable USB debugging, accept RSA, adb devices = device. Install the debug variant (com.safeher.app.debug).

STEP 18 -- Configure physical-device backend URL

1. Discover LAN IPv4 on the laptop.
2. Set DEBUG_API_BASE_URL=http://<LAN>:8000/ (trailing slash recommended; Gradle normalizes).
3. Rebuild/reinstall.
4. Start uvicorn with --host 0.0.0.0 --port 8000.
5. Confirm phone Wi-Fi (not mobile data only, unless you have a verified routing path -- NOT VERIFIED for cellular).

STEP 19 -- Test registration / login / profile

1. Register with a phone number + password (API POST /api/v1/auth/register).
2. Send OTP, then verify. With mock + shortcut, OTP 123456 is accepted (otp_service.py, OTP_ALLOW_DEV_SHORTCUT=true).
3. Login (POST /api/v1/auth/login).
4. Open Profile in the app.

Password rules: follow in-app validation (unit tests mention length checks). Exact UX copy: NOT VERIFIED beyond code.

STEP 20 -- Test SafeHer AI

1. Grant microphone + notifications.
2. Start AI / audio monitoring in the app (AudioMonitorService).
3. Confirm the foreground notification appears.
4. You should not get an automatic SOS (AI_AUTO_SOS_ENABLED=false). Confirm-first popup may appear on sustained distress; dismiss/confirm as designed.
5. Do not play loud distress audio in public without warning.

STEP 21 -- Test SOS flow

1. Grant location.
2. Add at least one emergency contact if the UI requires it for dispatch.
3. Use manual SOS if you need to exercise /api/v1/alerts/trigger.
4. Real WhatsApp delivery needs Meta credentials -- otherwise expect backend logs without a live message. Do not send a real emergency to real contacts.

STEP 22 -- Troubleshooting

See section 15.

15. Troubleshooting

Backend / Python

Table: Symptom / What to check

ModuleNotFoundError | Venv not activated; pip install -r requirements.txt not run; opened wrong folder
Wrong Python | python --version must be 3.11.x
pip / wheel failure | Use 3.11, upgrade pip, avoid 3.13 unless NOT VERIFIED
Postgres connection failure | Service running; DATABASE_URL user/password/db/port; alembic upgrade head; ping/ready
Firebase Admin errors | Missing JSON or bad FIREBASE_CREDENTIALS_PATH
Port 8000 in use | Another uvicorn; change port and URLs together

Android / Gradle

Table: Symptom / What to check

Gradle sync failure | Network for Maven; JDK; wrapper 9.3.0 download
Missing google-services.json | File must be exactly app/google-services.json
Android SDK missing | SDK Manager: Platform 35
sdk.dir missing | local.properties
TFLite missing | Tracked model should copy via syncTfliteAsset; check assets

Device / network

Table: Symptom / What to check

Device not detected | USB cable, drivers, adb kill-server then adb devices
unauthorized | Unlock phone, accept debugging RSA
offline | Replug, revoke USB debugging authorizations
Network error in app | Backend down; wrong base URL; cleartext blocked (should not happen on debug)
Emulator cannot reach API | Use 10.0.2.2 or LAN override + 0.0.0.0; not 127.0.0.1 inside emulator unless that is intended
Phone cannot reach API | 10.0.2.2 does not work on a physical phone. Use laptop LAN IP. Phone and PC on same Wi-Fi. Uvicorn 0.0.0.0. Windows Firewall inbound 8000. Disable guest-Wi-Fi AP isolation if the AP blocks client-to-client.
Windows Firewall | Allow Python/uvicorn TCP 8000 inbound on Private networks

10.0.2.2 = emulator alias for the host loopback.

Physical phone = developer laptop LAN IPv4.

Permissions / AI

Table: Symptom / What to check

No mic capture	RECORD_AUDIO denied; OEM permission drawer
No notification	POST_NOTIFICATIONS on API 33+
FGS killed	Battery optimization; FOREGROUND_SERVICE_MICROPHONE
TFLite load failure	Asset missing; Logcat TfliteAudioInferenceEngine
AI never auto-alerts	Expected: auto SOS is off

Logcat filter examples:

```
package:com.safeher.app.debug
tag:SAFEHER_AI
tag:okhttp.OkHttpClient
```

16. Auth, OTP, and CORS (local)

- Register/login: FastAPI JWT (SECRET_KEY, ALGORITHM=HS256).
- Mock OTP: stored code or 123456 when OTP_ALLOW_DEV_SHORTCUT=true.
- Production/staging refuse mock OTP (production_guards.py).
- Dev CORS: allow all origins.
- Do not enable AI_AUTO_SOS_ENABLED for setup.

17. Tests you can run locally

Backend (venv, Postgres not always required for every test -- many tests use the app with env from CI; NOT VERIFIED that every test passes without Postgres on every machine):

```
cd safeher_backend
.\.venv\Scripts\activate
pytest
```

CI equivalent env is in .github/workflows/ci.yml (DATABASE_URL to safeher_test, OTP_PROVIDER=mock, AI_AUTO_SOS_ENABLED=false).

Android:

```
cd safeher_android
.\gradlew.bat testDebugUnitTest
```

This guide does not claim a clean-machine installation was tested unless a later report says so.

18. Related documents

Table: Document / Use

docs/SAFEHER_DEVELOPER_SETUP.pdf	Printable copy of this guide
README.md	Short index
TEAM_HANDOVER.pdf	Older short handover
PRODUCTION_SETUP.md	Production DNS / hosting
safeher_backend/.env.example	Env names
safeher_ai/README.md	Retrain only

19. Verification notes (authoring)

Checked against repository files: Gradle versions, SDK levels, package name, .env.example keys, Alembic

filenames, uvicorn module path app.main:app, ping routes, OTP shortcut, gitignore entries, TFLite names, debug/release URLs, compose Postgres sample, CI Python 3.11.

Not executed as a blank-PC install. Android Studio exact version, OEM USB drivers, and live Firebase Console IAM were NOT VERIFIED.